

**STUDENT CASE STUDY EXECUTION REPORT**

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025 26)
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Course Outcome	CO-2

Case Study Topic: Elasticsearch Inverted Index — Multi-Term Query Intersection Optimization Using Merge/Galloping Search**Work Done Summary:**

In this case study, we simulated an Elasticsearch inverted index core mechanism by intersecting multiple sorted posting lists of document IDs. For multi-term queries (e.g., `machine AND learning AND tutorial`), computing the intersection efficiently is crucial for search engine response times.

We implemented a two-pointer merge-style intersection algorithm in Java. To optimize performance across multiple lists, we integrated a shortest-list-first strategy to minimize unnecessary comparisons. We executed a step-by-step trace intersecting three given posting lists (`list_machine`, `list_learning`, and `list_tutorial`) and analyzed the time complexity behavior.

Implementation Details:

- Two-Pointer Linear Merge:** Maintained pointers at the head of two sorted lists, advancing the pointer pointing to the smaller element, and recording matches only on equality.
- Shortest-List-First Strategy:** Sorted the query posting lists by length before beginning intersections to drop non-matching documents as early as possible.
- Multi-Way Evaluation:** Processed the intersection sequentially by taking the intermediate result of $(L1 \cap L2)$ and intersecting it with $L3$.
- Early Termination:** Stopped pointer traversals the moment any single list reached its end, minimizing redundant index scans.

Trace and Execution Steps:

Given Posting Lists:

- `list_machine` (L1 , size 8): [1, 4, 7, 12, 15, 20, 24, 30]
- `list_learning` (L2 , size 7): [2, 4, 8, 12, 18, 24, 30]
- `list_tutorial` (L3 , size 7): [4, 6, 9, 12, 17, 24, 30, 35]

Step 1: Intersecting L1 and L2 (`list_machine` $\cap$ `list_learning`)

- Pointer 1 (p1) tracks L1 , Pointer 2 (p2) tracks L2 .
- Compare 1 and 2 $\rightarrow$ p1 advances to 4.
- Compare 4 and 2 $\rightarrow$ p2 advances to 4.
- Compare 4 and 4 $\rightarrow$ Match found! Output: [4]. Both pointers advance.
- Compare 7 and 8 $\rightarrow$ p1 advances to 12.
- Compare 12 and 8 $\rightarrow$ p2 advances to 12.
- Compare 12 and 12 $\rightarrow$ Match found! Output: [4 , 12]. Both pointers advance.
- Compare 15 and 18 $\rightarrow$ p1 advances to 20.
- Compare 20 and 18 $\rightarrow$ p2 advances to 24.
- Compare 20 and 24 $\rightarrow$ p1 advances to 24.
- Compare 24 and 24 $\rightarrow$ Match found! Output: [4 , 12 , 24]. Both pointers advance.
- Compare 30 and 30 $\rightarrow$ Match found! Output: [4 , 12 , 24 , 30]. Both pointers advance.
- End of L2 reached.
- Intermediate Result (R1): [4 , 12 , 24 , 30]

Step 2: Intersecting R1 and L3 (`R1` $\cap$ `list_tutorial`)

- Pointer pR1 tracks R1 , Pointer p3 tracks L3 .
- Compare 4 and 4 $\rightarrow$ Match found! Output: [4]. Both advance.
- Compare 12 and 6 $\rightarrow$ p3 advances to 9.
- Compare 12 and 9 $\rightarrow$ p3 advances to 12.
- Compare 12 and 12 $\rightarrow$ Match found! Output: [4 , 12]. Both advance.
- Compare 24 and 17 $\rightarrow$ p3 advances to 24.
- Compare 24 and 24 $\rightarrow$ Match found! Output: [4 , 12 , 24]. Both advance.

- Compare 30 and 30 → Match found! Output: [4 , 12 , 24 , 30]. Both advance.
- End of R1 reached.

Result: Screenshots / Output

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Listening on 50082
User program running
User program finished
--- Elasticsearch Multi-Term Query Engine ---
Loading initial posting lists...
list_machine : [1, 4, 7, 12, 15, 20, 24, 30]
list_learning : [2, 4, 8, 12, 18, 24, 30]
list_tutorial : [4, 6, 9, 12, 17, 24, 30, 35]

[Optimization] Order of intersection (shortest first):
List 1 (Size 7): [2, 4, 8, 12, 18, 24, 30]
List 2 (Size 8): [1, 4, 7, 12, 15, 20, 24, 30]
List 3 (Size 8): [4, 6, 9, 12, 17, 24, 30, 35]

Executing Intersection step 1...
-> Intermediate Matches: [4, 12, 24, 30]
Executing Intersection step 2...
-> Intermediate Matches: [4, 12, 24, 30]

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FINAL INTERSECTED MATCHES (Document IDs): [4, 12, 24, 30]
Time Complexity (worst case):  $O(|L1| + |L2| + |L3|)$ 
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Process finished with exit code 0

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GitHub Link: https://github.com/kotamohith274/DSA-_2-Case-Studies.

Student Signature	Faculty Signature
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